

1.19 $\vec{J}' = \sigma \vec{E}'$ (Ohm's Law)

For NON-REL. VELOCITIES

$$\vec{E}' = \vec{E} + \vec{v} \times \vec{B}$$

$$\Rightarrow j^\mu = (pc, \vec{J}) = [pc, p\vec{v} + \sigma(\vec{E} + \vec{v} \times \vec{B})]$$

WHAT is the RELATIVISTIC GENERALIZATION?

$$(show \ j^\mu = \frac{1}{c^2} (u_\nu j^\nu) u^\mu + \sigma F^{\mu\nu} u_\nu)$$

Consider the non Relativistic Limit of

$$j^\mu = \frac{1}{c^2} (u_\nu j^\nu) u^\mu + \sigma F^{\mu\nu} u_\nu$$

$$= \frac{1}{c^2} [u_0 \gamma^0 - \vec{u} \cdot \vec{\gamma}] (\gamma^0 c, \gamma^0 \vec{v}) + \sigma [\frac{\vec{E}}{c} \cdot \vec{v},$$

$$= \frac{1}{c^2} [\gamma c p c - \vec{v} \cdot \vec{J}] ($$

$$\gamma c E_x - (E_y v_z - B_y v_z)$$

$$+ \gamma c E_y - (E_x v_z - B_x v_z)$$

$$+ \gamma c E_z - (E_x v_y - B_x v_y)$$

$$\gamma \vec{E} + \gamma \vec{v} \times \vec{B}]$$

$$j^\mu = [pc, \frac{\vec{v} \cdot \vec{J}}{c^2}] [c, \vec{v}] + \sigma [\frac{\vec{E}}{c} \cdot \vec{v}, \vec{E} + \vec{v} \times \vec{B}]$$

$$= [pc + \sigma \vec{E} \cdot \frac{\vec{v}}{c}, p\vec{v} + \sigma(\vec{E} + \vec{v} \times \vec{B})] \quad QED$$